

ELEC0447 – Analysis of Electrical Power and Energy Systems

Practical Sessions (TP 1 to 6) – Exercise Manual
Problem Statements and Step-by-Step Solutions

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Abstract

This document gathers the complete problem statements and rigorous, step-by-step solutions to the practical exercise sessions (TP 1 through TP 6) for the course **ELEC0447: Analysis of Electrical Power and Energy Systems**. It covers: phasor-domain circuit analysis and power factor correction (TP 1); three-phase circuits, power transfer between AC systems, radial voltage profiles, and the per-unit system (TP 2); transmission line Surge Impedance Loading (SIL), Joule losses, and distributed Telegrapher’s wave equations (TP 3); ideal and non-ideal transformer modeling with and without per-unit formalism (TP 4); synchronous generator steady-state/transient reactances, power-angle characteristics, and excitation control (TP 5); and voltage instability, analytical derivation of the P - V nose curve, maximum transmissible active power $P_{\max}$, critical voltage, and reactive power compensation effects (TP 6).

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1 Practical Session 1: Phasor-Domain Analysis

1.1 Exercises

1. Phasor Representation of Sinusoidal Voltages¹

Express the following voltages as phasors:

(a) $v_1(t) = \sqrt{2} \times 100 \cos(\omega t - 30^\circ) \text{ V}$

(b) $v_2(t) = \sqrt{2} \times 100 \cos(\omega t + 30^\circ) \text{ V}$

2. Series R - L - C Circuit

The series R - L - C circuit shown in Figure 1 is in sinusoidal steady state at a frequency of $f = 60 \text{ Hz}$. The parameters are:

$$V = 120 \text{ V}, \quad R = 1.5 \, \Omega, \quad L = 20 \text{ mH}, \quad C = 100 \, \mu\text{F}.$$

Calculate the current $i(t)$ in this circuit by using phasor-domain analysis.

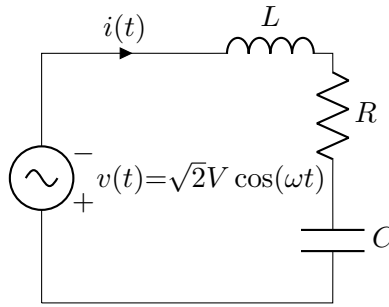


Figure 1: Series R - L - C circuit.

3. Series-Parallel R - L - C Circuit and Reactive Power Conservation

To the circuit of Figure 2, if a voltage of $100\angle 0^\circ \text{ V}$ is applied, calculate the active power P , the reactive power Q , and the power factor $\cos \phi$. Show that:

$$Q = \sum_k I_k^2 X_k$$

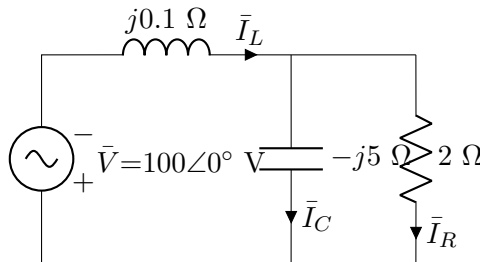


Figure 2: Series-parallel circuit for Exercise 3.

4. Power Factor Correction

In the circuit of Figure 3, the complex power drawn by the load impedance is:

$$S_L = P_L + jQ_L = (1858.4 + j1031.3) \text{ VA}.$$

Calculate the capacitive reactance X_C connected in parallel necessary to make the overall power factor equal to 0.9 (leading), assuming the applied voltage has an RMS value of $V = 120 \text{ V}$ at $f = 60 \text{ Hz}$. Determine the required capacitance C .

¹Exercises 2.1, 2.2, 2.5 and 2.9 from Ned Mohan's book "*Electric power systems, a first course*".

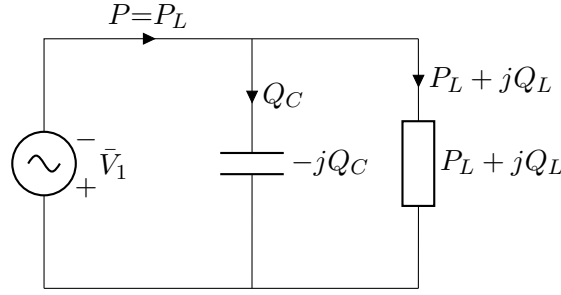


Figure 3: Power factor correction circuit.

1.2 Solutions

Solution to Exercise 1: Phasor Representation

Using the cosine reference with RMS value as phasor magnitude:

$$v(t) = \sqrt{2}V_{\text{RMS}} \cos(\omega t + \theta) \iff \bar{V} = V_{\text{RMS}}e^{j\theta} = V_{\text{RMS}}\angle\theta$$

(a) For $v_1(t) = \sqrt{2} \times 100 \cos(\omega t - 30^\circ)$ V:

$$\bar{V}_1 = V_{\text{RMS}}e^{j\theta} = \frac{V_m}{\sqrt{2}}e^{-j30^\circ} = 100 \text{ V} \angle -30^\circ \quad (1)$$

(b) For $v_2(t) = \sqrt{2} \times 100 \cos(\omega t + 30^\circ)$ V:

$$\bar{V}_2 = V_{\text{RMS}}e^{j\theta} = \frac{V_m}{\sqrt{2}}e^{+j30^\circ} = 100 \text{ V} \angle +30^\circ \quad (2)$$

Solution to Exercise 2: Series R - L - C Circuit

Data:

- Voltage: $\bar{V} = 120 \text{ V} \angle 0^\circ$
- Resistance: $R = 1.5 \Omega$
- Inductance: $L = 20 \text{ mH} = 20 \times 10^{-3} \text{ H}$
- Capacitance: $C = 100 \mu\text{F} = 100 \times 10^{-6} \text{ F}$
- Frequency: $f = 60 \text{ Hz} \implies \omega = 2\pi f = 2\pi \times 60 = 376.99 \text{ rad/s}$

Impedance Z : The total impedance of the series circuit is:

$$\begin{aligned} Z &= R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right) \\ &= 1.5 + j\left(376.99 \times 20 \times 10^{-3} - \frac{1}{376.99 \times 100 \times 10^{-6}}\right) \\ &= 1.5 + j(7.5398 - 26.5258) = 1.5 - j18.986 \Omega \\ &= 19.049 \Omega \angle -85.48^\circ \quad (-1.492 \text{ rad}) \end{aligned} \quad (3)$$

Current Phasor $\bar{I}$: By Ohm's law:

$$\bar{I} = \frac{\bar{V}}{Z} = \frac{120 \angle 0^\circ}{19.049 \angle -85.48^\circ} = 6.30 \text{ A} \angle +85.48^\circ \quad (+1.492 \text{ rad}) \quad (4)$$

Time-Domain Current $i(t)$: Converting back from the RMS phasor to the time domain:

$$\boxed{i(t) = 6.30\sqrt{2} \cos(376.99t + 1.492) \text{ A} = 8.91 \cos(376.99t + 85.5^\circ) \text{ A}} \quad (5)$$

Solution to Exercise 3: Series-Parallel Circuit and Power Conservation

Data:

- Applied voltage: $\bar{V} = 100 \text{ V} \angle 0^\circ$
- Inductor impedance: $Z_L = j0.1 \Omega$
- Capacitor impedance: $Z_C = -j5 \Omega$
- Resistor impedance: $R = 2 \Omega$

Total Equivalent Impedance Z : The capacitor and resistor are connected in parallel:

$$Z_p = Z_C \parallel R = \frac{R \cdot Z_C}{R + Z_C} = \frac{2(-j5)}{2 - j5} = \frac{-j10(2 + j5)}{2^2 + (-5)^2} = \frac{50 - j20}{29} = 1.7241 - j0.6897 \Omega \quad (6)$$

Adding the series inductor:

$$\begin{aligned} Z &= Z_L + Z_p = j0.1 + (1.7241 - j0.6897) = 1.7241 - j0.5897 \Omega \\ &= 1.822 \Omega \angle -18.88^\circ \end{aligned} \quad (7)$$

Complex Power S :

$$\begin{aligned} S &= \bar{V} \bar{I}^* = \bar{V} \left(\frac{\bar{V}}{Z} \right)^* = \frac{V^2}{Z^*} = \frac{100^2}{1.822 \angle +18.88^\circ} \\ &= 5487.93 \text{ VA} \angle -18.88^\circ \end{aligned} \quad (8)$$

Since $S = P + jQ$:

$$\boxed{P = \text{Re}\{S\} = 5487.93 \cos(-18.88^\circ) = 5192.54 \text{ W}} \quad (9)$$

$$\boxed{Q = \text{Im}\{S\} = 5487.93 \sin(-18.88^\circ) = -1775.89 \text{ var}} \quad (10)$$

Power Factor $\cos \phi$:

$$\boxed{\cos \phi = \frac{P}{|S|} = \frac{5192.54}{5487.93} = 0.946 \quad (\text{leading})} \quad (11)$$

Because $Q < 0$ (and $\phi < 0$), the circuit is capacitive overall: it produces net reactive power, meaning that the current phasor leads the voltage phasor.

Proof that $Q = \sum_k I_k^2 X_k$: We compute the branch currents:

- Current through the series inductor:

$$\bar{I}_L = \bar{I} = \frac{\bar{V}}{Z} = \frac{100 \angle 0^\circ}{1.822 \angle -18.88^\circ} = 54.88 \text{ A} \angle +18.88^\circ \implies I_L = 54.88 \text{ A}$$

- Current through the capacitor (using current divider):

$$\begin{aligned} \bar{I}_C &= \frac{R}{R + Z_C} \bar{I}_L = \frac{2}{2 - j5} (54.88 \angle 18.88^\circ) \\ &= \frac{2}{5.385 \angle -68.20^\circ} (54.88 \angle 18.88^\circ) = 20.38 \text{ A} \angle +87.08^\circ \implies I_C = 20.38 \text{ A} \end{aligned}$$

Calculating the individual reactive powers:

- Resistor: $Q_R = 0$ var
- Capacitor: $Q_C = X_C I_C^2 = (-5) \times (20.38)^2 = -2077.06$ var
- Inductor: $Q_L = X_L I_L^2 = 0.1 \times (54.88)^2 = +301.17$ var

Summing the individual contributions:

$$\sum_k I_k^2 X_k = Q_R + Q_C + Q_L = 0 - 2077.06 + 301.17 = -1775.89 \text{ var} = Q \quad (12)$$

This confirms that $Q = \sum_k I_k^2 X_k$. ■

Solution to Exercise 4: Power Factor Correction

Data:

- Load complex power: $S_L = P_L + jQ_L = (1858.4 + j1031.3)$ VA
- Applied voltage: $V = 120$ V (RMS)
- Desired overall power factor: $\cos \phi = 0.9$ (leading)
- Grid frequency: $f = 60$ Hz $\implies \omega = 2\pi f = 376.99$ rad/s

Determination of Total Reactive Power Q_{tot} : Connecting a pure capacitor in parallel does not alter the active power consumed by the system:

$$P_{\text{tot}} = P_L = 1858.4 \text{ W}$$

The new total complex power is $S = P_{\text{tot}} + jQ_{\text{tot}}$ with $Q_{\text{tot}} = Q_L + Q_C$. Using the power factor relationship:

$$\cos \phi = \frac{P_{\text{tot}}}{|S|} = \frac{P}{\sqrt{P^2 + Q_{\text{tot}}^2}} \iff Q_{\text{tot}}^2 = P^2 \left(\frac{1}{\cos^2 \phi} - 1 \right) \quad (13)$$

Taking the square root:

$$Q_{\text{tot}} = \pm P \sqrt{\frac{1}{\cos^2 \phi} - 1} = \pm 1858.4 \sqrt{\frac{1}{0.9^2} - 1} = \pm 900.064 \text{ var} \quad (14)$$

Because the corrected power factor is required to be **leading**, the total reactive power must be negative:

$$Q_{\text{tot}} = -900.064 \text{ var} \quad (15)$$

Capacitor Reactive Power Q_C : Since $Q_{\text{tot}} = Q_L + Q_C$:

$$Q_C = Q_{\text{tot}} - Q_L = -900.064 - 1031.3 = -1931.364 \text{ var} \quad (16)$$

Capacitive Reactance X_C : Across a voltage $V = 120$ V, the reactive power is related to the reactance by $Q_C = \frac{V^2}{X_C}$:

$$X_C = \frac{V^2}{Q_C} = \frac{120^2}{-1931.364} = -7.456 \text{ } \Omega \approx -7.46 \text{ } \Omega \quad (17)$$

Capacitance C : Since $X_C = -\frac{1}{\omega C}$:

$$C = \frac{1}{\omega |X_C|} = \frac{1}{2\pi \times 60 \times 7.456} = 3.557 \times 10^{-4} \text{ F} \approx 0.356 \text{ mF} = 356 \text{ } \mu\text{F} \quad (18)$$

2 Practical Session 2: Three-Phase Circuits and Power Transfer

2.1 Exercises

1. Balanced Three-Phase Voltage Source Phasors and Waveforms²

A positive sequence ($a-b-c$), balanced, wye-connected voltage source has the phase- a RMS phasor voltage given by:

$$\bar{V}_a = \sqrt{2} \times 100e^{j30^\circ} \text{ V.}$$

Obtain the time-domain voltages $v_a(t)$, $v_b(t)$, $v_c(t)$ and $v_{ab}(t)$, and show all of these as phasors in a phasor diagram.

2. Balanced Three-Phase Inductive Load

A balanced three-phase inductive load is supplied in steady state by a balanced, wye-connected, three-phase voltage source with a phase voltage of 120 V RMS. The load draws a total active power of 10 kW at a power factor of 0.9 (inductive).

- Calculate the RMS value of the phase currents.
- Calculate the magnitude and phase of the per-phase load impedance, assuming a wye-connected load.
- Draw a phasor diagram showing all three phase voltages and currents.

3. Power Transfer Between Two AC Systems

In the per-phase circuit of Figure 4, the active power transfer per phase is $P = 1$ kW from side S (sending) to side R (receiving). The parameters are:

$$V_S = 100 \text{ V}, \quad \bar{V}_R = 95\angle 0^\circ \text{ V}, \quad X = 1.5 \Omega.$$

Calculate the line current $\bar{I}$, the phase angle θ_S of $\bar{V}_S$, and the per-phase reactive power Q_R supplied to the receiving end.

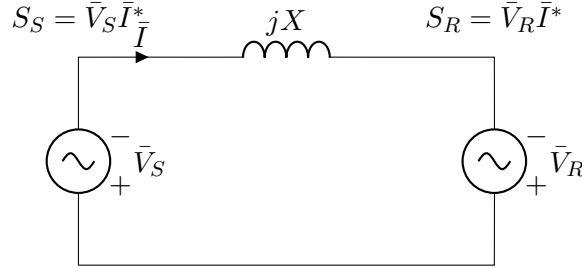


Figure 4: Per-phase model of power transfer between two AC systems.

4. Voltage Profile of a Radial Transmission System

In a radial system represented by the circuit of Figure 4 where the receiving end is a passive load, the series reactance is $X = 1.5 \Omega$. Consider the sending-end voltage to be held constant at $\bar{V}_S = 100\angle 0^\circ \text{ V}$.

- Derive a formula for Q_R , the reactive power consumed by the load, in terms of P_R and the power factor $\cos \phi$.
- Derive a second-order polynomial equation for V_R^2 that depends on X , V_S , P_R , and Q_R .
- Calculate and discuss the ratio V_S/V_R and the power angle δ when the active load P_R varies from 0 to 1 kW for three power factors:

²Exercises 2.11, 2.12, 2.16, 2.17, 2.18, 2.19 and 2.20 from Ned Mohan's book "*Electric power systems, a first course*".

- Unity ($\cos \phi = 1.0$)
- Inductive ($\cos \phi = 0.9$ lagging)
- Capacitive ($\cos \phi = 0.9$ leading)

5. Per-Unit Analysis of a Three-Phase System

In the balanced three-phase circuit of Figure 5, the load impedance magnitude is $|Z_L| = 10 \Omega$ per phase, and the per-phase power factor is 0.8 (lagging). Calculate the base values, the per-unit values of the load impedance $Z_{L,pu}$, and the load real and reactive powers $P_{1\phi,pu}$ and $Q_{1\phi,pu}$ for each of the following base specifications:

- Line-to-line voltage base $U^B = 208 \text{ V}$, three-phase power base $S_{3\phi}^B = 3.6 \text{ kVA}$.
- Line-to-line voltage base $U^B = 240 \text{ V}$, three-phase power base $S_{3\phi}^B = 5.4 \text{ kVA}$.
- Line-to-line voltage base $U^B = 240 \text{ V}$, three-phase power base $S_{3\phi}^B = 3.6 \text{ kVA}$.

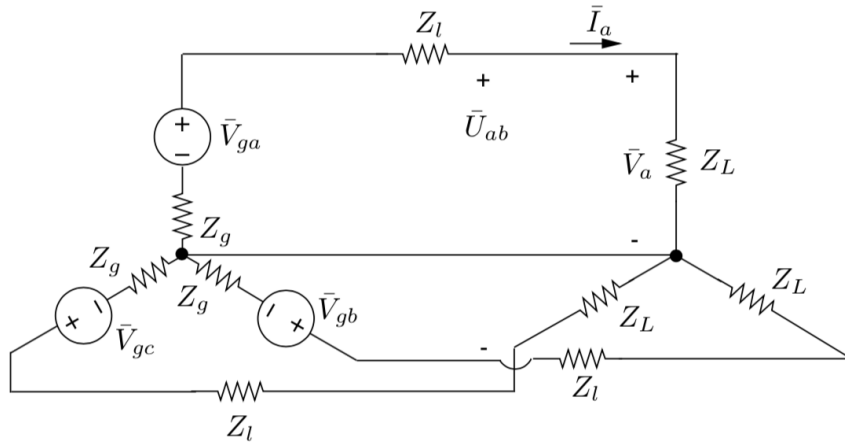


Figure 5: Balanced wye-connected three-phase source and load system.

2.2 Solutions

Solution to Exercise 1: Three-Phase Source Phasors and Waveforms

Data: Positive sequence (a - b - c) balanced source with:

$$\bar{V}_a = \sqrt{2} \times 100e^{j30^\circ} \text{ V} = 141.42 \text{ V} \angle 30^\circ = 122.47 + j70.71 \text{ V}$$

Since the phasor is given in RMS, the amplitude of the time-domain sinusoidal wave is $\sqrt{2} \times |\bar{V}_a| = \sqrt{2} \times 141.42 = 200 \text{ V}$.

Time-Domain Voltages:

$$\boxed{v_a(t) = 200 \cos(\omega t + 30^\circ) \text{ V}} \quad (19)$$

In a balanced positive-sequence system, phase b lags phase a by 120° , and phase c leads phase a by 120° (or lags by 240°):

$$\begin{aligned} \bar{V}_b &= \bar{V}_a e^{-j120^\circ} = \sqrt{2} \times 100e^{j(30^\circ - 120^\circ)} \\ &= \sqrt{2} \times 100e^{-j90^\circ} = -j141.42 \text{ V} \end{aligned} \quad (20)$$

$$\boxed{v_b(t) = 200 \cos(\omega t - 90^\circ) \text{ V}} \quad (21)$$

$$\begin{aligned} \bar{V}_c &= \bar{V}_a e^{+j120^\circ} = \sqrt{2} \times 100e^{j(30^\circ + 120^\circ)} \\ &= \sqrt{2} \times 100e^{j150^\circ} = -122.47 + j70.71 \text{ V} \end{aligned} \quad (22)$$

$$\boxed{v_c(t) = 200 \cos(\omega t + 150^\circ) \text{ V}} \quad (23)$$

Line-to-Line Voltage $\bar{V}_{ab}$:

$$\begin{aligned} \bar{V}_{ab} &= \bar{V}_a - \bar{V}_b = \sqrt{2} \times 100 \left(e^{j30^\circ} - e^{-j90^\circ} \right) \\ &= (122.47 + j70.71) - (-j141.42) = 122.47 + j212.13 \text{ V} \\ &= 244.95 \text{ V} \angle 60^\circ = \sqrt{3} \bar{V}_a e^{j30^\circ} \end{aligned} \quad (24)$$

The peak amplitude is $200\sqrt{3} = 346.41 \text{ V}$:

$$\boxed{v_{ab}(t) = 200\sqrt{3} \cos(\omega t + 60^\circ) \text{ V} = 346.41 \cos(\omega t + 60^\circ) \text{ V}} \quad (25)$$

Solution to Exercise 2: Balanced Three-Phase Inductive Load

Data:

- Phase voltage: $V_{1\phi} = 120 \text{ V RMS}$
- Total 3-phase active power: $P_{3\phi} = 10 \text{ kW} = 10^4 \text{ W}$
- Power factor: $\cos \phi = 0.9$ (inductive/lagging) $\implies \phi = +\arccos(0.9) = +25.842^\circ$

1. Phase Current RMS Value: The per-phase active power is:

$$P_{1\phi} = \frac{P_{3\phi}}{3} = \frac{10^4}{3} = 3333.33 \text{ W}$$

Using $P_{1\phi} = V_{1\phi} I_{\text{RMS}} \cos \phi$:

$$\boxed{I_{\text{RMS}} = \frac{P_{1\phi}}{V_{1\phi} \cos \phi} = \frac{3333.33}{120 \times 0.9} = 30.86 \text{ A}} \quad (26)$$

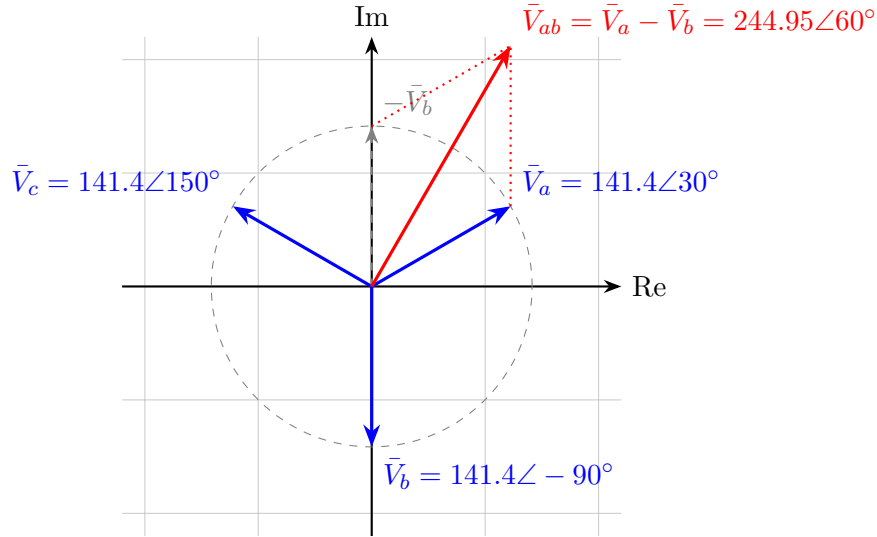


Figure 6: Phasor diagram of balanced phase voltages and line-to-line voltage $\bar{V}_{ab}$.

2. Load Impedance $Z_{1\phi}$: For a wye connection, the voltage across each phase impedance is the phase voltage $V_{1\phi} = 120$ V:

$$|Z_{1\phi}| = \frac{V_{1\phi}}{I_{\text{RMS}}} = \frac{120}{30.864} = 3.888 \, \Omega \quad (27)$$

Since the load is inductive, current lags voltage by $\phi = 25.84^\circ$, meaning the impedance angle is positive:

$$\boxed{Z_{1\phi} = 3.888 \, \Omega \angle + 25.84^\circ = (3.50 + j1.70) \, \Omega} \quad (28)$$

3. Phasor Diagram: Taking $\bar{V}_a = 120 \angle 0^\circ$ V as reference:

- Voltages: $\bar{V}_a = 120 \angle 0^\circ$ V, $\bar{V}_b = 120 \angle -120^\circ$ V, $\bar{V}_c = 120 \angle +120^\circ$ V
- Currents: $\bar{I}_a = 30.86 \, \text{A} \angle -25.84^\circ$, $\bar{I}_b = 30.86 \, \text{A} \angle -145.84^\circ$, $\bar{I}_c = 30.86 \, \text{A} \angle +94.16^\circ$

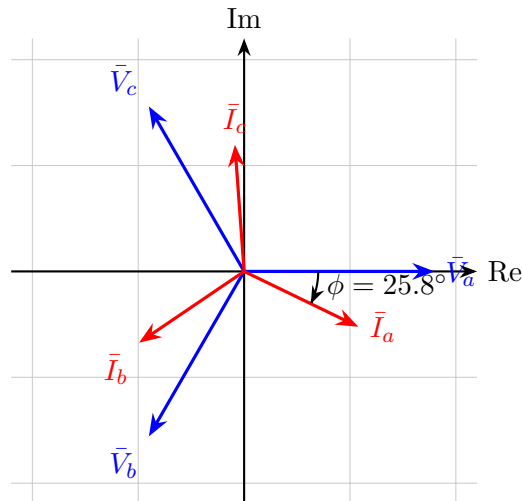


Figure 7: Phasor diagram of voltages and lagging currents for the balanced inductive load.

Solution to Exercise 3: Power Transfer Between Two AC Systems

Data:

- Transferred active power: $P = 1 \text{ kW} = 1000 \text{ W}$
- Sending-end voltage: $V_S = 100 \text{ V}$ (angle θ_S to be determined)
- Receiving-end voltage: $\bar{V}_R = 95 \angle 0^\circ \implies V_R = 95 \text{ V}, \theta_R = 0^\circ$
- Line reactance: $X = 1.5 \Omega$ (lossless line, $R = 0$)

Derivation of Power Transfer Formulas: The line current directed from S to R is:

$$\bar{I} = \frac{\bar{V}_S - \bar{V}_R}{jX} \quad (29)$$

The complex power injected by the sending-end source S is:

$$\begin{aligned} S_S &= \bar{V}_S \bar{I}^* = \bar{V}_S \left(\frac{\bar{V}_S - \bar{V}_R}{jX} \right)^* = \frac{V_S^2 - \bar{V}_S \bar{V}_R^*}{-jX} = j \frac{V_S^2 - V_S V_R e^{j(\theta_S - \theta_R)}}{X} \\ &= \frac{V_S V_R \sin(\theta_S - \theta_R)}{X} + j \frac{V_S^2 - V_S V_R \cos(\theta_S - \theta_R)}{X} \end{aligned} \quad (30)$$

Hence:

$$P_S = \frac{V_S V_R \sin(\theta_S - \theta_R)}{X} \quad (31)$$

$$Q_S = \frac{V_S^2 - V_S V_R \cos(\theta_S - \theta_R)}{X} \quad (32)$$

Phase Angle θ_S : Since the line is lossless ($R = 0$), $P_S = P_R = 1000 \text{ W}$. With $\theta_R = 0^\circ$:

$$\sin \theta_S = \frac{X P_S}{V_S V_R} = \frac{1.5 \times 1000}{100 \times 95} = \frac{1500}{9500} = 0.15789 \quad (33)$$

$$\boxed{\theta_S = \arcsin(0.15789) = 9.085^\circ \quad (0.1585 \text{ rad})} \quad (34)$$

Line Current $\bar{I}$:

$$\begin{aligned} \bar{I} &= \frac{100 \angle 9.085^\circ - 95 \angle 0^\circ}{j1.5} = \frac{(98.743 + j15.789) - 95}{j1.5} \\ &= \frac{3.743 + j15.789}{j1.5} = \frac{15.789 - j3.743}{1.5} = 10.526 - j2.495 \text{ A} \\ &= \boxed{10.818 \text{ A} \angle -13.35^\circ} \end{aligned} \quad (35)$$

Per-Phase Reactive Power Q_R Supplied to the Receiving End: The power received at bus R is:

$$\begin{aligned} S_R &= \bar{V}_R \bar{I}^* = (95 \angle 0^\circ)(10.526 + j2.495) \\ &= 999.97 + j237.06 \text{ VA} \approx 1000 \text{ W} + j237.22 \text{ var} \end{aligned} \quad (36)$$

Alternatively, using the analytical receiving-end formula:

$$\begin{aligned} Q_R &= \text{Im}\{\bar{V}_R \bar{I}^*\} = \frac{V_S V_R \cos(\theta_R - \theta_S) - V_R^2}{X} \\ &= \frac{100 \times 95 \times \cos(-9.085^\circ) - 95^2}{1.5} \\ &= \frac{9500 \times 0.9874 - 9025}{1.5} = \frac{9380.57 - 9025}{1.5} = \boxed{237.05 \text{ var} \approx 237.22 \text{ var}} \end{aligned} \quad (37)$$

Note on sign convention: As annotated in the handwritten notes, $Q_R > 0$ indicates that the line supplies 237.22 var of reactive power into the receiving-end source/bus.

Solution to Exercise 4: Voltage Profile of a Radial Transmission System

(a) **Load Reactive Power Q_R :** Let the load complex power be $S_R = P_R + jQ_R$. By definition of the load power factor:

$$\cos \phi = \frac{P_R}{|S_R|} = \frac{P_R}{\sqrt{P_R^2 + Q_R^2}} \iff Q_R = \pm P_R \sqrt{\frac{1}{\cos^2 \phi} - 1} \quad (38)$$

- **Unity power factor ($\cos \phi = 1$):** $Q_R = 0$.
- **Lagging power factor ($\cos \phi = 0.9$):** $Q_R = +P_R \sqrt{\frac{1}{0.9^2} - 1} = +0.4843P_R > 0$ (inductive load consumes reactive power).
- **Leading power factor ($\cos \phi = 0.9$):** $Q_R = -P_R \sqrt{\frac{1}{0.9^2} - 1} = -0.4843P_R < 0$ (capacitive load generates reactive power).

(b) **Second-Order Equation for V_R^2 :** Let $\delta = \theta_R - \theta_S$ be the transmission angle. The power flow equations at the receiving end are:

$$P_R = \frac{V_S V_R \sin \delta}{X} \implies V_R \sin \delta = \frac{P_R X}{V_S} \quad (39)$$

$$Q_R = \frac{V_S V_R \cos \delta - V_R^2}{X} \implies V_R \cos \delta = \frac{X Q_R + V_R^2}{V_S} \quad (40)$$

Squaring and summing equations (39) and (40):

$$V_R^2 (\sin^2 \delta + \cos^2 \delta) = V_R^2 = \left(\frac{P_R X}{V_S} \right)^2 + \left(\frac{X Q_R + V_R^2}{V_S} \right)^2 \quad (41)$$

Multiplying both sides by V_S^2 :

$$V_S^2 V_R^2 = P_R^2 X^2 + X^2 Q_R^2 + 2X Q_R V_R^2 + V_R^4 \quad (42)$$

Rearranging terms into standard polynomial form in $y = V_R^2$:

$$\boxed{y^2 + (2X Q_R - V_S^2)y + X^2(P_R^2 + Q_R^2) = 0 \quad \text{with} \quad y = V_R^2} \quad (43)$$

(c) **Ratio V_S/V_R and Power Angle δ :** Solving the quadratic equation for $y = V_R^2$:

$$V_R^2 = \frac{-(2X Q_R - V_S^2) + \sqrt{(2X Q_R - V_S^2)^2 - 4X^2(P_R^2 + Q_R^2)}}{2} \quad (44)$$

(choosing the positive square root corresponding to normal, stable operating voltage). Once V_R is obtained, the power angle δ is:

$$\delta = \arcsin \left(\frac{P_R X}{V_S V_R} \right) \quad (45)$$

Physical Interpretation:

- **Lagging power factor (inductive):** As P_R increases, the line reactive drop adds to the load reactive demand, causing a severe drop in receiving voltage V_R , so $V_S/V_R > 1$ increases rapidly.
- **Unity power factor:** V_R exhibits a moderate voltage drop due to the line series reactance.
- **Leading power factor (capacitive):** The capacitive reactive power supports the voltage (Ferranti-like compensation effect), keeping V_R elevated ($V_S/V_R \approx 1$ or even < 1 for light loads).

Solution to Exercise 5: Per-Unit Analysis of a Three-Phase System

Data:

- Impedance magnitude: $|Z_L| = 10 \Omega$
- Power factor: $\cos \phi = 0.8$ lagging $\implies \phi = \arccos(0.8) = +36.87^\circ$
- Load impedance: $Z_L = 10 \Omega \angle 36.87^\circ = (8 + j6) \Omega$

Per-Unit Formulas:

- Phase voltage base: $V^B = \frac{U^B}{\sqrt{3}}$
- Single-phase power base: $S_{1\phi}^B = \frac{S_{3\phi}^B}{3}$
- Current base: $I^B = \frac{S_{1\phi}^B}{V^B} = \frac{S_{3\phi}^B}{\sqrt{3}U^B}$
- Base impedance: $Z^B = \frac{(V^B)^2}{S_{1\phi}^B} = \frac{(U^B)^2}{S_{3\phi}^B}$

Table 1: Base quantities and per-unit values for Exercise 5.

Quantity	Case (a)	Case (b)	Case (c)
U^B (line-to-line)	208 V	240 V	240 V
$S_{3\phi}^B$	3.6 kVA	5.4 kVA	3.6 kVA
$V^B = U^B/\sqrt{3}$	120.09 V	138.56 V	138.56 V
$S_{1\phi}^B = S_{3\phi}^B/3$	1.2 kVA	1.8 kVA	1.2 kVA
$I^B = S_{1\phi}^B/V^B$	10.0 A	13.0 A	8.66 A
$Z^B = (U^B)^2/S_{3\phi}^B$	12.018 Ω	10.667 Ω	16.000 Ω
$ Z_{L,pu} = Z_L /Z^B$	0.832 pu	0.9375 pu	0.625 pu
$Z_{L,pu}$	0.832$\angle 36.87^\circ$ pu	0.938$\angle 36.87^\circ$ pu	0.625$\angle 36.87^\circ$ pu
$ S_{1\phi} = (V^B)^2/ Z_L $	1442.13 VA	1920.0 VA	1920.0 VA
$P_{1\phi} = S_{1\phi} \cos \phi$	1153.71 W	1536.0 W	1536.0 W
$Q_{1\phi} = S_{1\phi} \sin \phi$	865.28 var	1152.0 var	1152.0 var
$P_{pu} = P_{1\phi}/S_{1\phi}^B$	0.9614 pu	0.8533 pu	1.2800 pu
$Q_{pu} = Q_{1\phi}/S_{1\phi}^B$	0.7213 pu	0.6400 pu	0.9600 pu

Detailed Calculations for the Three Base Systems: Notice that in per-unit:

$$S_{pu} = \frac{|V_{pu}|^2}{Z_{L,pu}^*} = \frac{1}{Z_{L,pu}^*} = \frac{1}{|Z_{L,pu}|} \angle 36.87^\circ$$

For example, in case (a): $|S_{pu}| = 1/0.832 = 1.2018$ pu, giving $P_{pu} = 1.2018 \times 0.8 = 0.9614$ pu and $Q_{pu} = 1.2018 \times 0.6 = 0.7211$ pu, which matches perfectly.

3 Practical Session 3: The Transmission Line

3.1 Exercises

1. Surge Impedance Loading (SIL) and Line Losses³

The parameters for a 500-kV transmission system with bundled conductors are as follows:

$$Z_c = 258 \, \Omega, \quad R = 1.76 \times 10^{-2} \, \Omega/\text{km}.$$

- Demonstrate why the voltage magnitude along a lossless line terminated in its surge impedance Z_c is constant.
- Calculate the Surge Impedance Loading (SIL) of the transmission system.
- Calculate the percentage active power loss in this transmission line if it is 300 km long and operates loaded at its SIL.

2. Distributed-Parameter Line Model and Voltage Profiles

Table 2 gives approximate transmission line parameters with bundled conductors at 60 Hz.

Table 2: Approximate transmission line parameters with bundled conductors at 60 Hz.

Nominal voltage [kV]	R [Ω/km]	ωL [Ω/km]	ωC [$\mu\text{S}/\text{km}$]
230	0.06	0.50	3.4
345	0.04	0.38	4.6
500	0.03	0.33	5.3
765	0.01	0.34	5.0

A 200 km long 345-kV line has the parameters given in Table 2. Neglect the line series resistance ($R = 0$).

- Starting from an infinitesimal section of length dx (Figure 8), derive the Telegrapher's wave equations in the time domain, and then convert them into the phasor domain.
- Solve the differential equations in terms of the receiving-end boundary conditions $\bar{V}_R$ and $\bar{I}_R$.
- Calculate the voltage profile along the line if the receiving end is held at 1.0 pu ($V_R = 345$ kV) with $\cos \phi = 1.0$, for:
 - A load of $1.5 \times \text{SIL}$
 - A load of $0.75 \times \text{SIL}$

Evaluate the voltage magnitude at $x = 100$ km and at the sending end $x = 200$ km.

³Exercises 4.3 and 4.10 of Ned Mohan's book "*Electric power systems, a first course*".

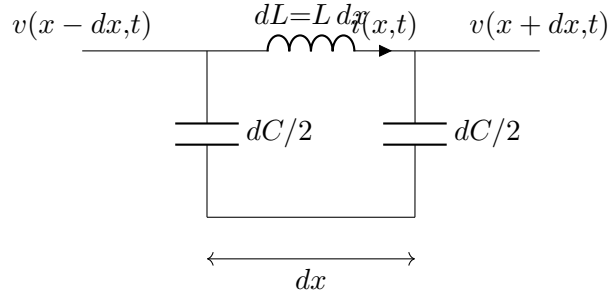


Figure 8: Infinitesimal section dx of a lossless distributed transmission line.

3.2 Solutions

Solution to Exercise 1: Surge Impedance Loading (SIL) and Losses

Data:

- Rated line-to-line voltage: $V_{LL} = 500$ kV
- Surge impedance: $Z_c = 258 \Omega$
- Resistance per unit length: $R = 1.76 \times 10^{-2} \Omega/\text{km}$
- Line length: $l = 300$ km

1. Voltage Flatness at SIL Termination: The general expression for the voltage phasor at a distance x from the receiving end on a lossless transmission line is:

$$\bar{V}(x) = \bar{V}_R \cos(\beta x) + jZ_c \bar{I}_R \sin(\beta x) \quad (46)$$

When the line is terminated by its characteristic impedance Z_c under a resistive load ($\bar{V}_R = V_R \angle 0^\circ$ and $\cos \phi = 1$), the receiving-end current is:

$$\bar{I}_R = \frac{\bar{V}_R}{Z_c} \quad (47)$$

Substituting this into (46):

$$\bar{V}(x) = \bar{V}_R \cos(\beta x) + jZ_c \left(\frac{\bar{V}_R}{Z_c} \right) \sin(\beta x) = \bar{V}_R (\cos(\beta x) + j \sin(\beta x)) = \bar{V}_R e^{j\beta x} \quad (48)$$

Taking the magnitude:

$$|\bar{V}(x)| = |\bar{V}_R| = \text{constant} \quad \forall x \quad (49)$$

Thus, the voltage magnitude remains strictly constant along the entire length of the line: only the phase angle advances linearly with x by βx .

2. Surge Impedance Loading (SIL): By definition, the Surge Impedance Loading is the total three-phase active power delivered into a load equal to the surge impedance Z_c at nominal voltage:

$$\text{SIL} = \frac{V_{LL}^2}{Z_c} = \frac{(500 \times 10^3)^2}{258} = 9.6899 \times 10^8 \text{ W} \approx 969 \text{ MW} \quad (50)$$

3. Percentage Power Loss for $l = 300$ km: At rated SIL, the equivalent line current is:

$$I = \frac{\text{SIL}}{V_{LL}} = \frac{968.99 \times 10^6 \text{ W}}{500 \times 10^3 \text{ V}} = 1937.98 \text{ A} \quad (51)$$

The total resistance of the line is:

$$R_{\text{tot}} = R \cdot l = (1.76 \times 10^{-2} \Omega/\text{km}) \times 300 \text{ km} = 5.28 \Omega \quad (52)$$

The Joule losses in the line are:

$$P_{\text{loss}} = R_{\text{tot}} I^2 = 5.28 \times (1937.98)^2 = 19.831 \times 10^6 \text{ W} = 19.831 \text{ MW} \quad (53)$$

The relative percentage loss is:

$$\boxed{\% \text{ loss} = \frac{P_{\text{loss}}}{\text{SIL}} \times 100 = \frac{19.831 \text{ MW}}{968.99 \text{ MW}} \times 100 = 2.05\%} \quad (54)$$

Solution to Exercise 2: Distributed Line Model and Voltage Profile

Data for 345-kV Line:

- Rated voltage: $V_{LL} = 345 \text{ kV} = 3.45 \times 10^5 \text{ V}$
- Length: $l = 200 \text{ km}$
- Series reactance per unit length: $\omega L = 0.38 \Omega/\text{km}$
- Shunt susceptance per unit length: $\omega C = 4.6 \mu\text{S}/\text{km} = 4.6 \times 10^{-6} \text{ S}/\text{km}$
- Line resistance: $R = 0$ (lossless line assumption)

Characteristic Parameters:

- Surge impedance:

$$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{\omega L}{\omega C}} = \sqrt{\frac{0.38}{4.6 \times 10^{-6}}} = 287.417 \Omega \quad (55)$$

- Phase propagation constant:

$$\beta = \omega \sqrt{LC} = \sqrt{(\omega L)(\omega C)} = \sqrt{0.38 \times 4.6 \times 10^{-6}} = 1.322 \times 10^{-3} \text{ rad/km} = 0.001322 \text{ rad/km} \quad (56)$$

- Surge Impedance Loading:

$$\text{SIL} = \frac{V_{LL}^2}{Z_c} = \frac{(345 \times 10^3)^2}{287.417} = 4.1412 \times 10^8 \text{ VA} = 414.12 \text{ MVA (MW)} \quad (57)$$

1. Derivation of Telegrapher's Equations: Consider an infinitesimal line slice of length dx with distributed inductance L and capacitance C per unit length. Applying Kirchhoff's Current Law (KCL):

$$\begin{aligned} i(x - dx, t) &= i(x, t) + (C dx) \frac{\partial v(x, t)}{\partial t} \\ \Leftrightarrow \frac{i(x, t) - i(x - dx, t)}{dx} &= -C \frac{\partial v(x, t)}{\partial t} \end{aligned} \quad (58)$$

Taking the limit as $dx \rightarrow 0$:

$$-\frac{\partial i(x, t)}{\partial x} = C \frac{\partial v(x, t)}{\partial t} \quad (59)$$

Applying Kirchhoff's Voltage Law (KVL):

$$\begin{aligned} v(x, t) &= (L dx) \frac{\partial i(x, t)}{\partial t} + v(x + dx, t) \\ \Longleftrightarrow \frac{v(x + dx, t) - v(x, t)}{dx} &= -L \frac{\partial i(x, t)}{\partial t} \end{aligned} \quad (60)$$

Taking the limit as $dx \rightarrow 0$:

$$-\frac{\partial v(x, t)}{\partial x} = L \frac{\partial i(x, t)}{\partial t} \quad (61)$$

Equations (59) and (61) are the celebrated **Telegrapher's equations** in the time domain. Differentiating (59) with respect to x and (61) with respect to t :

$$-\frac{\partial^2 i}{\partial x^2} = C \frac{\partial^2 v}{\partial x \partial t} \quad (62)$$

$$-\frac{\partial^2 v}{\partial t \partial x} = L \frac{\partial^2 i}{\partial t^2} \quad (63)$$

Substituting:

$$\boxed{\frac{\partial^2 i(x, t)}{\partial x^2} = LC \frac{\partial^2 i(x, t)}{\partial t^2}} \quad (64)$$

Similarly for the voltage:

$$\boxed{\frac{\partial^2 v(x, t)}{\partial x^2} = LC \frac{\partial^2 v(x, t)}{\partial t^2}} \quad (65)$$

Phasor Domain Representation: Under sinusoidal steady state, $\frac{\partial}{\partial t} \rightarrow j\omega$ and $\frac{\partial^2}{\partial t^2} \rightarrow -\omega^2$. Equation (64) becomes:

$$\frac{d^2 \bar{I}(x)}{dx^2} + \omega^2 LC \bar{I}(x) = 0 \Longleftrightarrow \frac{d^2 \bar{I}(x)}{dx^2} + \beta^2 \bar{I}(x) = 0 \quad (66)$$

where $\beta = \omega\sqrt{LC}$. The general solution of this linear second-order differential equation is:

$$\bar{I}(x) = I_1 e^{j\beta x} + I_2 e^{-j\beta x} \quad (67)$$

From (59) in the phasor domain:

$$\begin{aligned} -\frac{d\bar{I}(x)}{dx} &= j\omega C \bar{V}(x) \implies \bar{V}(x) = -\frac{1}{j\omega C} \frac{d\bar{I}(x)}{dx} \\ &= -\frac{1}{j\omega C} (j\beta I_1 e^{j\beta x} - j\beta I_2 e^{-j\beta x}) \\ &= -\frac{\beta}{\omega C} (I_1 e^{j\beta x} - I_2 e^{-j\beta x}) \end{aligned} \quad (68)$$

Since $\frac{\beta}{\omega C} = \frac{\omega\sqrt{LC}}{\omega C} = \sqrt{\frac{L}{C}} = Z_c$:

$$\bar{V}(x) = Z_c (-I_1 e^{j\beta x} + I_2 e^{-j\beta x}) \quad (69)$$

2. Boundary Conditions at Receiving End ($x = 0$): At the receiving end $x = 0$, let $\bar{V}(0) = \bar{V}_R$ and $\bar{I}(0) = \bar{I}_R$:

$$\begin{cases} \bar{V}_R = Z_c (-I_1 + I_2) \\ \bar{I}_R = I_1 + I_2 \end{cases} \Longleftrightarrow \begin{cases} -I_1 + I_2 = \frac{\bar{V}_R}{Z_c} \\ I_1 + I_2 = \bar{I}_R \end{cases} \quad (70)$$

Solving for I_1 and I_2 :

$$I_1 = \frac{\bar{I}_R - \bar{V}_R/Z_c}{2} \quad (71)$$

$$I_2 = \frac{\bar{I}_R + \bar{V}_R/Z_c}{2} \quad (72)$$

Substituting (71) and (72) into (69):

$$\begin{aligned} \bar{V}(x) &= Z_c \left[- \left(\frac{\bar{I}_R - \bar{V}_R/Z_c}{2} \right) e^{j\beta x} + \left(\frac{\bar{I}_R + \bar{V}_R/Z_c}{2} \right) e^{-j\beta x} \right] \\ &= \bar{V}_R \left(\frac{e^{j\beta x} + e^{-j\beta x}}{2} \right) - Z_c \bar{I}_R \left(\frac{e^{j\beta x} - e^{-j\beta x}}{2} \right) \\ &= \boxed{\bar{V}(x) = \bar{V}_R \cos(\beta x) + j Z_c \bar{I}_R \sin(\beta x)} \end{aligned} \quad (73)$$

and similarly:

$$\boxed{\bar{I}(x) = \bar{I}_R \cos(\beta x) + j \frac{\bar{V}_R}{Z_c} \sin(\beta x)} \quad (74)$$

3. Voltage Profiles for Loading Conditions: At the receiving end, the voltage is held at nominal 1 pu:

$$\bar{V}_R = 3.45 \times 10^5 \angle 0^\circ \text{ V.}$$

The load operates at unity power factor ($\cos \phi = 1.0$), so $\bar{I}_R = I_R \angle 0^\circ$.

(a) **Heavy loading ($1.5 \times \text{SIL}$):**

$$P_L = 1.5 \times 414.12 \text{ MW} = 621.18 \text{ MW}$$

$$\bar{I}_R = \frac{P_L}{V_R} = \frac{621.18 \times 10^6 \text{ W}}{3.45 \times 10^5 \text{ V}} = 1800.518 \text{ A} \angle 0^\circ$$

(b) **Light loading ($0.75 \times \text{SIL}$):**

$$P_L = 0.75 \times 414.12 \text{ MW} = 312.059 \text{ MW}$$

$$\bar{I}_R = \frac{P_L}{V_R} = \frac{312.059 \times 10^6 \text{ W}}{3.45 \times 10^5 \text{ V}} = 900.259 \text{ A} \angle 0^\circ$$

Numerical Evaluation along the Line (for Case a: $1.5 \times \text{SIL}$):

- At $x = 0$ (receiving end):

$$\bar{V}(0) = \bar{V}_R = 345.0 \text{ kV} \angle 0^\circ \quad (1.000 \text{ pu})$$

- At $x = 100 \text{ km}$:

$$\beta x = (0.001322 \text{ rad/km}) \times 100 \text{ km} = 0.1322 \text{ rad} = 7.574^\circ$$

$$\cos(\beta x) = \cos(0.1322 \text{ rad}) = 0.99128, \quad \sin(\beta x) = \sin(0.1322 \text{ rad}) = 0.13182$$

Substituting into (73):

$$\begin{aligned} \bar{V}(100 \text{ km}) &= (3.45 \times 10^5)(0.99128) + j(287.417 \times 1800.518)(0.13182) \\ &= 341.99 \times 10^3 + j68.22 \times 10^3 \text{ V} \\ &= 3.4872 \times 10^5 \text{ V} \angle 11.28^\circ \quad (1.0108 \text{ pu}) \end{aligned} \quad (75)$$

$$\boxed{|\bar{V}(100 \text{ km})| \approx 3.49 \times 10^5 \text{ V} = 349 \text{ kV}} \quad (76)$$

- At $x = 200$ km (sending end):

$$\beta x = (0.001322 \text{ rad/km}) \times 200 \text{ km} = 0.2644 \text{ rad} = 15.149^\circ$$

$$\cos(\beta x) = 0.96525, \quad \sin(\beta x) = 0.26133$$

$$\begin{aligned} \bar{V}(200 \text{ km}) &= (3.45 \times 10^5)(0.96525) + j(287.417 \times 1800.518)(0.26133) \\ &= 333.01 \times 10^3 + j135.24 \times 10^3 \text{ V} \\ &= 3.5944 \times 10^5 \text{ V} \angle 22.10^\circ \quad (1.0419 \text{ pu}) \end{aligned} \tag{77}$$

$$\boxed{|\bar{V}(200 \text{ km})| = 359.44 \text{ kV} = 1.042 \text{ pu}} \tag{78}$$

Physical Interpretation of Voltage Profiles:

- When loaded above SIL ($P > \text{SIL}$): The line's series inductive reactive power consumption exceeds the reactive power generated by its shunt capacitance. The sending-end voltage must be higher than the receiving-end voltage ($V_S > V_R$) to supply this deficit.
- When loaded below SIL ($P < \text{SIL}$): The capacitive reactive power generated by the line exceeds the inductive absorption. Consequently, voltage rises along the line towards the receiving end (the **Ferranti effect**), meaning $V_R > V_S$ if not properly regulated with shunt reactors.
- At exact SIL ($P = \text{SIL}$): Shunt capacitive generation exactly balances series inductive consumption ($Q_{\text{gen}} = Q_{\text{cons}}$). The voltage magnitude profile is perfectly flat along the line.

4 Practical Session 4: Transformers in Power Systems

4.1 Exercises

1. Ideal Transformer Circuit⁴

Assume the transformer in Figure 9 to be ideal, with a turns ratio $N_1/N_2 = 3$. Winding 1 is connected to a sinusoidal voltage source in steady state with:

$$\bar{V}_1 = 120 \text{ V} \angle 0^\circ \quad \text{at } f = 60 \text{ Hz.}$$

The load on winding 2 is a series combination of resistance R and inductive reactance X , with impedance:

$$Z_L = (5 + j3) \Omega.$$

Calculate the current $\bar{I}_1$ drawn from the voltage source.

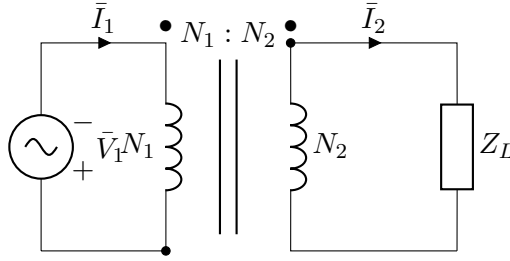


Figure 9: Ideal transformer with secondary load.

2. Real Transformer Equivalent Circuit and Per-Unit Formalism

A 2400/240-V, 60-Hz single-phase transformer has the following parameters in its equivalent circuit (Figure 10):

- High-side (primary) leakage impedance: $Z_{t1} = R_{t1} + jX_{t1} = (1.2 + j2.0) \Omega$
- Low-side (secondary) leakage impedance: $Z_{t2} = R_{t2} + jX_{t2} = (0.012 + j0.02) \Omega$
- Core loss resistance R_{fe} is neglected ($R_{fe} \rightarrow \infty$).

The output voltage at the secondary terminals is held at $V_2 = 240 \text{ V}$ (RMS), supplying a load of impedance magnitude $|Z_L| = 1.5 \Omega$ at a power factor of 0.9 (lagging).

Calculate the primary input voltage $\bar{V}_1$:

- If the magnetizing reactance at the high side is $X_m = 1800 \Omega$.
- If the magnetizing reactance X_m is neglected ($X_m \rightarrow \infty$).

Solve the problem using two methods:

- **Method 1:** In actual physical units (without the per-unit formalism).
- **Method 2:** Using the per-unit formalism, considering the (2400 V, 38400 VA) base on the primary side and the (240 V, 38400 VA) base on the secondary side.

⁴Exercises 6.2 and 6.4 from Ned Mohan's book "Electric power systems, a first course".

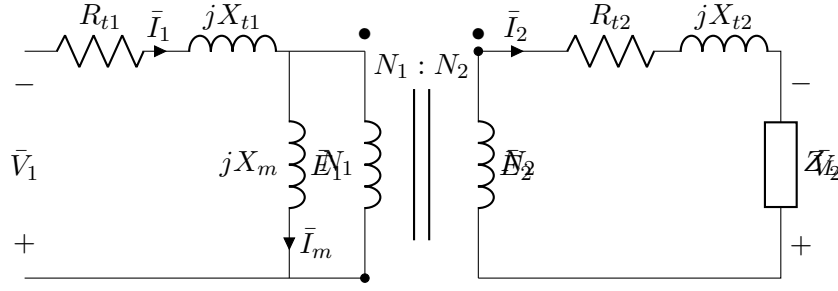


Figure 10: Transformer equivalent circuit neglecting core loss resistance R_{fe} .

4.2 Solutions

Solution to Exercise 1: Ideal Transformer

Data:

- Turns ratio: $n = \frac{N_2}{N_1} = \frac{1}{3} \implies \frac{N_1}{N_2} = 3$
- Primary voltage: $\bar{V}_1 = 120 \text{ V} \angle 0^\circ$
- Load impedance: $Z_L = (5 + j3) \Omega = 5.831 \Omega \angle 30.96^\circ$

Solution Scheme: We propagate through the system: $\bar{V}_1 \rightarrow \bar{V}_2 \rightarrow \bar{I}_2 \rightarrow \bar{I}_1$.

1. **Secondary Voltage $\bar{V}_2$:** For an ideal transformer, the voltage ratio is proportional to the turns ratio:

$$\bar{V}_2 = \frac{N_2}{N_1} \bar{V}_1 = \frac{1}{3} \times 120 \angle 0^\circ = 40 \text{ V} \angle 0^\circ \quad (79)$$

2. **Secondary Current $\bar{I}_2$:** Applying Ohm's law to the secondary load:

$$\bar{I}_2 = \frac{\bar{V}_2}{Z_L} = \frac{40 \angle 0^\circ}{5 + j3} = \frac{40 \angle 0^\circ}{5.831 \angle 30.96^\circ} = 6.86 \text{ A} \angle -30.96^\circ \approx 6.86 \text{ A} \angle -31^\circ \quad (80)$$

3. **Primary Current $\bar{I}_1$:** The amp-turn balance for an ideal transformer ($N_1 \bar{I}_1 = N_2 \bar{I}_2$) yields:

$$\bar{I}_1 = \frac{N_2}{N_1} \bar{I}_2 = \frac{1}{3} \times (6.86 \angle -30.96^\circ) = 2.29 \text{ A} \angle -30.96^\circ \approx 2.29 \text{ A} \angle -31^\circ \quad (81)$$

Alternative Verification by Impedance Reflection: The secondary impedance reflected to the primary is:

$$Z'_L = \left(\frac{N_1}{N_2} \right)^2 Z_L = 3^2 \times (5 + j3) = 45 + j27 \Omega = 52.48 \Omega \angle 30.96^\circ$$

The primary current directly obtained from the primary side is:

$$\bar{I}_1 = \frac{\bar{V}_1}{Z'_L} = \frac{120 \angle 0^\circ}{52.48 \angle 30.96^\circ} = 2.29 \text{ A} \angle -30.96^\circ$$

Solution to Exercise 2: Real Transformer

Data:

- Rated voltages: $V_{\text{rated},1} = 2400 \text{ V}$, $V_{\text{rated},2} = 240 \text{ V} \implies n = \frac{N_2}{N_1} = \frac{240}{2400} = \frac{1}{10}$
- Primary leakage impedance: $Z_{t1} = 1.2 + j2.0 \Omega$
- Secondary leakage impedance: $Z_{t2} = 0.012 + j0.02 \Omega$
- Secondary load: $|Z_L| = 1.5 \Omega$, $\cos \phi = 0.9$ lagging $\implies \phi = \arccos(0.9) = 25.842^\circ$
- Secondary terminal voltage: $\bar{V}_2 = 240 \text{ V} \angle 0^\circ$

Part 1: Solution Without the Per-Unit Formalism

1. Secondary Current $\bar{I}_2$:

$$\begin{aligned}\bar{I}_2 &= \frac{\bar{V}_2}{|Z_L| \angle \phi} = \frac{240 \angle 0^\circ}{1.5 \angle 25.842^\circ} = 160 \text{ A} \angle -25.842^\circ \\ &= 160(\cos 25.842^\circ - j \sin 25.842^\circ) = 144 - j69.742 \text{ A}\end{aligned}\quad (82)$$

2. Internal Induced Voltage $\bar{E}_2$:

$$\begin{aligned}\bar{E}_2 &= \bar{V}_2 + \bar{I}_2(R_{t2} + jX_{t2}) \\ &= 240 + (144 - j69.742)(0.012 + j0.02) \\ &= 240 + (1.728 + j2.88 - j0.8369 + 1.3948) \\ &= 243.123 + j2.043 \text{ V} = \mathbf{243.131 \text{ V} \angle 0.4815^\circ}\end{aligned}\quad (83)$$

3. Primary Induced Voltage $\bar{E}_1$: Through the ideal transformer ratio $\frac{N_1}{N_2} = 10$:

$$\bar{E}_1 = \frac{N_1}{N_2} \bar{E}_2 = 10 \times 243.131 \angle 0.4815^\circ = \mathbf{2431.31 \text{ V} \angle 0.4815^\circ} = (2431.23 + j20.43) \text{ V} \quad (84)$$

4. Magnetizing Current $\bar{I}_m$:

- Case (a) with $X_m = 1800 \Omega$:

$$\begin{aligned}\bar{I}_m &= \frac{\bar{E}_1}{jX_m} = \frac{2431.31 \angle 0.4815^\circ}{1800 \angle 90^\circ} \\ &= \mathbf{1.351 \text{ A} \angle -89.519^\circ} = (0.0113 - j1.3507) \text{ A}\end{aligned}\quad (85)$$

- Case (b) with $X_m \rightarrow \infty$:

$$\bar{I}_m = 0 \text{ A} \quad (86)$$

5. Referred Secondary Current $\bar{I}'_2$:

$$\bar{I}'_2 = \frac{N_2}{N_1} \bar{I}_2 = \frac{160 \angle -25.842^\circ}{10} = \mathbf{16 \text{ A} \angle -25.842^\circ} = (14.4 - j6.9742) \text{ A} \quad (87)$$

6. Primary Current $\bar{I}_1 = \bar{I}'_2 + \bar{I}_m$:

- Case (a):

$$\begin{aligned}\bar{I}_1 &= (14.4 - j6.9742) + (0.0113 - j1.3507) = 14.4113 - j8.3249 \text{ A} \\ &= \mathbf{16.643 \text{ A} \angle -30.013^\circ}\end{aligned}\quad (88)$$

- **Case (b):**

$$\bar{I}_1 = \bar{I}'_2 = \mathbf{16\ A\angle -25.842^\circ} \quad (89)$$

7. **Primary Voltage** $\bar{V}_1 = \bar{E}_1 + \bar{I}_1(R_{t1} + jX_{t1})$:

- **Case (a):**

$$\begin{aligned} \bar{I}_1 Z_{t1} &= (14.4113 - j8.3249)(1.2 + j2.0) = 33.943 + j18.833\ \text{V} \\ \bar{V}_1 &= (2431.23 + j20.43) + (33.943 + j18.833) = 2465.17 + j39.263\ \text{V} \\ &= \boxed{\mathbf{2465.48\ V\angle 0.9125^\circ}} \end{aligned} \quad (90)$$

- **Case (b):**

$$\begin{aligned} \bar{I}'_2 Z_{t1} &= (14.4 - j6.9742)(1.2 + j2.0) = 31.228 + j20.431\ \text{V} \\ \bar{V}_1 &= (2431.23 + j20.43) + (31.228 + j20.431) = 2462.46 + j40.861\ \text{V} \\ &= \boxed{\mathbf{2462.80\ V\angle 0.9507^\circ}} \end{aligned} \quad (91)$$

Part 2: Solution With the Per-Unit Formalism

1. Base Quantities: We select:

- Common apparent power base: $S_{\text{base}} = 38400 \text{ VA}$
- Primary voltage base: $V_{\text{base1}} = 2400 \text{ V}$
- Secondary voltage base: $V_{\text{base2}} = 240 \text{ V}$

Derived base quantities:

$$Z_{\text{base1}} = \frac{V_{\text{base1}}^2}{S_{\text{base}}} = \frac{2400^2}{38400} = 150 \text{ } \Omega, \quad I_{\text{base1}} = \frac{S_{\text{base}}}{V_{\text{base1}}} = \frac{38400}{2400} = 16 \text{ A} \quad (92)$$

$$Z_{\text{base2}} = \frac{V_{\text{base2}}^2}{S_{\text{base}}} = \frac{240^2}{38400} = 1.5 \text{ } \Omega, \quad I_{\text{base2}} = \frac{S_{\text{base}}}{V_{\text{base2}}} = \frac{38400}{240} = 160 \text{ A} \quad (93)$$

2. Parameters in Per-Unit:

$$Z_{t1,pu} = \frac{R_{t1} + jX_{t1}}{Z_{\text{base1}}} = \frac{1.2 + j2.0}{150} = \mathbf{0.008 + j0.01333} \text{ pu} \quad (94)$$

$$Z_{t2,pu} = \frac{R_{t2} + jX_{t2}}{Z_{\text{base2}}} = \frac{0.012 + j0.02}{1.5} = \mathbf{0.008 + j0.01333} \text{ pu} \quad (95)$$

Crucial Per-Unit Property: The per-unit impedances of both windings are identical:

$$Z_{t1,pu} = Z_{t2,pu}$$

The magnetizing reactance in per-unit is:

$$X_{m,pu} = \frac{X_m}{Z_{\text{base1}}} = \frac{1800}{150} = 12 \text{ pu} \quad (96)$$

The load impedance and terminal voltage in per-unit:

$$|Z_{L,pu}| = \frac{|Z_L|}{Z_{\text{base2}}} = \frac{1.5}{1.5} = 1.0 \text{ pu} \implies Z_{L,pu} = 1 \angle 25.842^\circ \text{ pu} \quad (97)$$

$$\bar{V}_{2,pu} = \frac{\bar{V}_2}{V_{\text{base2}}} = \frac{240 \angle 0^\circ}{240} = 1 \angle 0^\circ \text{ pu} \quad (98)$$

3. Secondary Current $\bar{I}_{2,pu}$:

$$\bar{I}_{2,pu} = \frac{\bar{V}_{2,pu}}{Z_{L,pu}} = \frac{1 \angle 0^\circ}{1 \angle 25.842^\circ} = \mathbf{1 \text{ pu} \angle -25.842^\circ} \quad (99)$$

4. Induced Voltages $\bar{E}_{2,pu}$ and $\bar{E}_{1,pu}$:

$$\begin{aligned} \bar{E}_{2,pu} &= \bar{V}_{2,pu} + \bar{I}_{2,pu} Z_{t2,pu} \\ &= 1 \angle 0^\circ + (1 \angle -25.842^\circ)(0.008 + j0.01333) = \mathbf{1.0130 \text{ pu} \angle 0.4815^\circ} \end{aligned} \quad (100)$$

Because the per-unit voltage bases are matched to the turns ratio ($V_{\text{base1}}/V_{\text{base2}} = N_1/N_2$), the ideal transformer has a ratio of 1 : 1 in per-unit:

$$\bar{E}_{1,pu} = \bar{E}_{2,pu} = \mathbf{1.0130 \text{ pu} \angle 0.4815^\circ} \quad (101)$$

5. Magnetizing Current $\bar{I}_{m,pu}$:

- Case (a):

$$\bar{I}_{m,pu} = \frac{\bar{E}_{1,pu}}{jX_{m,pu}} = \frac{1.0130 \angle 0.4815^\circ}{j12} = \mathbf{0.0844 \text{ pu} \angle -89.519^\circ} \quad (102)$$

- Case (b):

$$\bar{I}_{m,pu} = 0 \text{ pu} \quad (103)$$

6. Primary Current $\bar{I}_{1,pu}$:

- Case (a):

$$\bar{I}_{1,pu} = \bar{I}_{2,pu} + \bar{I}_{m,pu} = (1 \angle -25.842^\circ) + (0.0844 \angle -89.519^\circ) = \mathbf{1.04019 \text{ pu} \angle -30.0135^\circ} \quad (104)$$

- Case (b):

$$\bar{I}_{1,pu} = \bar{I}_{2,pu} = \mathbf{1 \text{ pu} \angle -25.842^\circ} \quad (105)$$

7. Primary Voltage $\bar{V}_{1,pu}$ and Conversion to Volts:

- Case (a):

$$\bar{V}_{1,pu} = \bar{E}_{1,pu} + \bar{I}_{1,pu} Z_{t1,pu} = \mathbf{1.0273 \text{ pu} \angle 0.9125^\circ} \quad (106)$$

$$\bar{V}_1 = \bar{V}_{1,pu} \times V_{\text{base1}} = 1.0273 \angle 0.9125^\circ \times 2400 \text{ V} = \boxed{\mathbf{2465.48 \text{ V} \angle 0.9125^\circ}} \quad (107)$$

- Case (b): When magnetizing reactance is neglected, the entire transformer collapses to a single total series impedance:

$$Z_{tr,pu} = Z_{t1,pu} + Z_{t2,pu} = 2Z_{t1,pu} = (0.016 + j0.0267) \text{ pu} \quad (108)$$

Therefore:

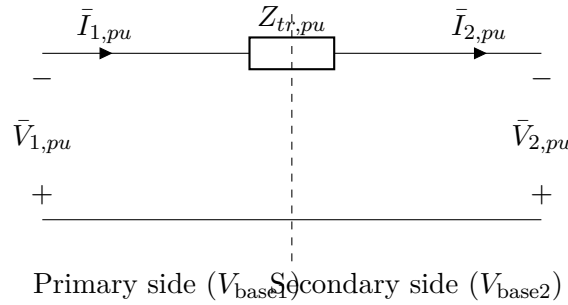


Figure 11: Simplified per-unit equivalent circuit of the transformer with $X_m \rightarrow \infty$.

$$\begin{aligned} \bar{V}_{1,pu} &= \bar{V}_{2,pu} + Z_{tr,pu} \bar{I}_{2,pu} \\ &= 1 \angle 0^\circ + (0.016 + j0.0267)(1 \angle -25.842^\circ) = 1.026 + j0.0170 = \mathbf{1.0262 \text{ pu} \angle 0.9507^\circ} \end{aligned} \quad (109)$$

$$\bar{V}_1 = \bar{V}_{1,pu} \times V_{\text{base1}} = 1.0262 \angle 0.9507^\circ \times 2400 \text{ V} = \boxed{\mathbf{2462.80 \text{ V} \angle 0.9507^\circ}} \quad (110)$$

Both methods yield identical results to 5 significant digits, demonstrating the power and compactness of the per-unit formulation.

5 Practical Session 5: Synchronous Generators in Power Systems

5.1 Exercises

1. Per-Phase Equivalent Circuit and Power Limits⁵

In the per-phase equivalent circuit of Figure 12, assume $R_s = 0$ and $X_s = 1.2$ pu. The terminal voltage is $\bar{V}_a = 1\angle 0^\circ$ pu and the armature current is $\bar{I}_a = 1\angle -\pi/6$ pu.

- Calculate the internal field excitation voltage phasor $\bar{E}_{af}$.
- Show that the three-phase active power produced at the stator is equivalent to the three-phase active power transferred between two voltage sources ($\bar{E}_{af}$ and $\bar{V}_\infty$) where $\bar{V}_\infty = 1\angle 0^\circ$ pu is an ideal voltage source.
- If E_{af} is held constant, calculate the maximum active power this machine can deliver in theory and in practice.

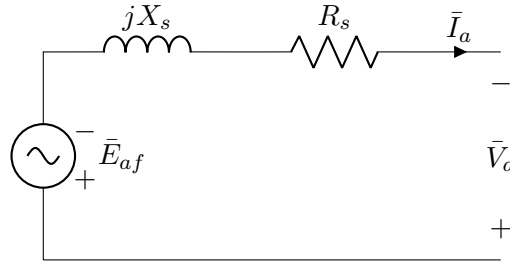


Figure 12: Per-phase equivalent circuit of a synchronous generator.

2. Reactive Power Control and $E_{af} \sin \delta$ Invariance

In a synchronous generator, assume $R_s = 0$ and $X_s = 1.2$ pu. The terminal voltage is $\bar{V}_a = 1\angle 0^\circ$ pu. The machine supplies a constant three-phase active power of $P_{3\phi} = 1.0$ pu.

- Calculate $\bar{I}_a$ such that the active power remains constant while the three-phase reactive power takes the values:
 - $Q_{3\phi} = 0$ pu
 - $Q_{3\phi} = 0.5$ pu
 - $Q_{3\phi} = -0.5$ pu
- For each configuration, show that $E_{af} \sin \delta$ remains constant.
- Sketch and explain the phasor diagrams corresponding to overexcited and underexcited operation.

3. Subtransient, Transient, and Steady-State Regimes

In a synchronous generator, assume $R_s = 0$. The steady-state, transient, and subtransient reactances are respectively given by:

$$X_s = 1.2 \text{ pu}, \quad X'_s = 0.33 \text{ pu}, \quad X''_s = 0.23 \text{ pu}.$$

The terminal voltage is $\bar{V}_a = 1\angle 0^\circ$ pu and the current is $\bar{I}_a = 1\angle -\pi/6$ pu.

- Calculate the internal voltage phasor $\bar{E}_{af}$ in each configuration: steady-state, transient, and subtransient.
- Compare and explain the physical origin of these reactances.

⁵Exercises 9.5, 9.6, 9.7 and 9.9 of Ned Mohan's book "*Electric power systems, a first course*".

5.2 Solutions

Solution to Exercise 1: Generator Per-Phase Model and Power Limits

Data:

- Armature resistance: $R_s = 0 \Omega$
- Synchronous reactance: $X_s = 1.2 \text{ pu}$
- Terminal voltage: $\bar{V}_a = 1 \angle 0^\circ \text{ pu} = 1 + j0 \text{ pu}$
- Stator current: $\bar{I}_a = 1 \angle -\pi/6 \text{ pu} = 1 \angle -30^\circ \text{ pu} = \left(\frac{\sqrt{3}}{2} - j\frac{1}{2}\right) \text{ pu}$

(a) **Internal Excitation Voltage $\bar{E}_{af}$:** Applying Kirchhoff's voltage law to the stator circuit:

$$\begin{aligned}
 \bar{E}_{af} &= \bar{V}_a + R_s \bar{I}_a + jX_s \bar{I}_a \\
 &= 1 + 0 + j1.2 \left(\frac{\sqrt{3}}{2} - j\frac{1}{2} \right) \\
 &= 1 + 0.6 + j1.2 \frac{\sqrt{3}}{2} = 1.6 + j1.0392 \text{ pu} \\
 &= \boxed{1.908 \text{ pu} \angle 33.0^\circ \quad (0.576 \text{ rad} \approx 0.58 \text{ rad})}
 \end{aligned} \tag{111}$$

Thus:

$$|E_{af}| = 1.908 \text{ pu}, \quad \delta = 33.0^\circ$$

(b) **Active Power Equivalence:**

1. Active power transferred between two voltage sources ($\bar{E}_{af} = |E_{af}| \angle \delta$ and $\bar{V}_\infty = |V_a| \angle 0^\circ$):

$$\begin{aligned}
 P_{3\phi}^{2VS} &= 3 \times \frac{|E_{af}| |V_a| \sin \delta}{X_s} \\
 &= 3 \times \frac{1.908 \times 1.0 \times \sin(33.0^\circ)}{1.2} \\
 &= 3 \times \frac{1.908 \times 0.5446}{1.2} = \mathbf{2.598 \text{ pu}} \approx \mathbf{2.60 \text{ pu}}
 \end{aligned} \tag{112}$$

Note: As marked in the original notes, the factor 3 accounts for the total three-phase power across all three phases.

2. Three-phase active power produced at the stator:

$$\begin{aligned}
 P_{3\phi}^{\text{stator}} &= \text{Re}\{3\bar{V}_a \bar{I}_a^*\} = 3|V_a| |I_a| \cos(\angle \bar{V}_a - \angle \bar{I}_a) \\
 &= 3 \times 1.0 \times 1.0 \times \cos(0^\circ - (-30^\circ)) \\
 &= 3 \cos(30^\circ) = 3 \times \frac{\sqrt{3}}{2} = \mathbf{2.598 \text{ pu}} \approx \mathbf{2.60 \text{ pu}}
 \end{aligned} \tag{113}$$

Hence:

$$P_{3\phi}^{2VS} = P_{3\phi}^{\text{stator}} \blacksquare \tag{114}$$

(c) **Maximum Active Power Machine Can Deliver:** The power-angle curve is given by:

$$P_{3\phi}(\delta) = 3 \frac{|E_{af}||V_a|}{X_s} \sin \delta \quad (115)$$

- **Theoretical Maximum Power** ($\delta = 90^\circ = \pi/2$): Differentiating with respect to δ :

$$\frac{\partial P_{3\phi}}{\partial \delta} = 3 \frac{|E_{af}||V_a|}{X_s} \cos \delta = 0 \implies \delta = 90^\circ$$

$$P_{\max, \text{th}} = 3 \frac{|E_{af}||V_a|}{X_s} = 3 \times \frac{1.908 \times 1.0}{1.2} = 4.770 \text{ pu} \quad (116)$$

- **Practical Operating Limit** ($\delta \leq 40^\circ$): In power system operations, a steady-state stability margin is maintained to withstand transient disturbances without losing synchronism. Typical practical limits restrict the rotor angle to $\delta \leq 40^\circ$:

$$P_{\max, \text{pr}} = 3 \frac{|E_{af}||V_a| \sin(40^\circ)}{X_s} = 4.770 \times \sin(40^\circ) = 3.066 \text{ pu} \quad (117)$$

Solution to Exercise 2: Reactive Power Control and $E_{af} \sin \delta$ Invariance

Data:

- Constant three-phase active power: $P_{3\phi} = 1.0 \text{ pu}$
- Terminal voltage: $\bar{V}_a = 1 \angle 0^\circ \text{ pu}$
- Reactance: $X_s = 1.2 \text{ pu}$, $R_s = 0$

1. **Armature Current $\bar{I}_a$ for each $Q_{3\phi}$:** The total three-phase complex power is:

$$S_{3\phi} = 3\bar{V}_a\bar{I}_a^* = P_{3\phi} + jQ_{3\phi} \implies \bar{I}_a = \left(\frac{P_{3\phi} + jQ_{3\phi}}{3\bar{V}_a} \right)^* = \frac{P_{3\phi} - jQ_{3\phi}}{3V_a}$$

(a) **For $Q_{3\phi} = 0 \text{ pu}$ (Unity Power Factor):**

$$\bar{I}_a = \frac{1 - j0}{3} = \frac{1}{3} \text{ pu} = 0.333 \text{ pu} \angle 0^\circ \quad (118)$$

(b) **For $Q_{3\phi} = +0.5 \text{ pu}$ (Lagging Power Factor):**

$$\bar{I}_a = \frac{1 - j0.5}{3} = 0.3333 - j0.1667 \text{ pu} \quad (119)$$

$$\bar{I}_a = 0.3727 \text{ pu} \angle -26.565^\circ \quad (-0.464 \text{ rad}) \quad (120)$$

(c) **For $Q_{3\phi} = -0.5 \text{ pu}$ (Leading Power Factor):**

$$\bar{I}_a = \frac{1 - (-j0.5)}{3} = 0.3333 + j0.1667 \text{ pu} \quad (121)$$

$$\bar{I}_a = 0.3727 \text{ pu} \angle +26.565^\circ \quad (+0.464 \text{ rad}) \quad (122)$$

2. Demonstration that $E_{af} \sin \delta = \text{constant}$: From the active power equation:

$$P_{3\phi} = 3 \frac{|E_{af}| |V_a| \sin \delta}{X_s} \quad (123)$$

Rearranging for $|E_{af}| \sin \delta$:

$$|E_{af}| \sin \delta = \frac{P_{3\phi} X_s}{3 |V_a|} = \frac{1.0 \times 1.2}{3 \times 1.0} = 0.4 \text{ pu} = \text{constant} \quad (124)$$

Similarly, for the armature current:

$$I_a \cos \phi = \frac{P_{3\phi}}{3 |V_a|} = \frac{1.0}{3 \times 1.0} = \frac{1}{3} \text{ pu} = \text{constant} \quad (125)$$

This geometrically means that as Q varies at constant P , the tip of the voltage phasor $\bar{E}_{af}$ moves along a horizontal line at vertical height 0.4 pu above the real axis, and the current phasor $\bar{I}_a$ moves along a vertical line at horizontal distance $1/3$ pu.

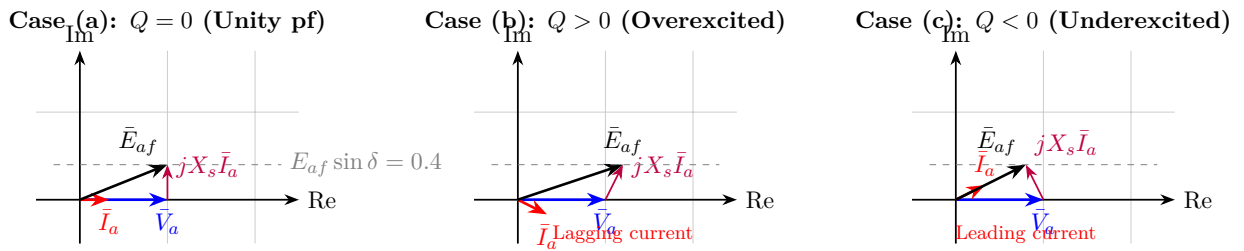


Figure 13: Phasor diagrams of synchronous generator operating at constant active power P under variable excitation.

3. Phasor Diagrams and Operating Modes:

Physical Explanation of Generator Voltage Control:

- **Overexcitation (Case b):** By increasing the rotor field current I_f , the internal EMF magnitude $|E_{af}|$ increases. Because P is held constant, δ must decrease while $\bar{I}_a$ shifts into the lagging quadrant. The generator delivers reactive power to the grid ($Q > 0$), effectively acting as a **shunt capacitor**.
- **Underexcitation (Case c):** By decreasing the field current I_f , $|E_{af}|$ decreases. The current shifts into the leading quadrant, and the generator absorbs reactive power from the grid ($Q < 0$), acting as a **shunt inductor**.

This capability to continuously adjust reactive power by varying the field current forms the foundation of **Automatic Voltage Regulation (AVR)** in bulk power transmission networks.

Solution to Exercise 3: Subtransient, Transient, and Steady-State Regimes

Data:

- Armature resistance: $R_s = 0$
- Steady-state synchronous reactance: $X_s = 1.2$ pu
- Transient reactance: $X'_s = 0.33$ pu
- Subtransient reactance: $X''_s = 0.23$ pu
- Terminal voltage: $\bar{V}_a = 1 \angle 0^\circ$ pu
- Current: $\bar{I}_a = 1 \angle -\pi/6$ pu $= 1 \angle -30^\circ$ pu $= (0.866 - j0.5)$ pu

1. Internal Voltages in each Regime: Using $\bar{E} = \bar{V}_a + jX\bar{I}_a$:

(a) **Steady-State Voltage $\bar{E}_{af}$:**

$$\begin{aligned}\bar{E}_{af} &= 1 + j1.2(0.866 - j0.5) = 1 + 0.6 + j1.0392 = 1.6 + j1.0392 \text{ pu} \\ &= \boxed{1.908 \text{ pu} \angle 33.0^\circ \quad (0.58 \text{ rad})}\end{aligned}\tag{126}$$

(b) **Transient Voltage $\bar{E}'_{af}$:**

$$\begin{aligned}\bar{E}'_{af} &= 1 + j0.33(0.866 - j0.5) = 1 + 0.165 + j0.2858 = 1.165 + j0.2858 \text{ pu} \\ &= \boxed{1.200 \text{ pu} \angle 13.78^\circ \quad (0.24 \text{ rad})}\end{aligned}\tag{127}$$

(c) **Subtransient Voltage $\bar{E}''_{af}$:**

$$\begin{aligned}\bar{E}''_{af} &= 1 + j0.23(0.866 - j0.5) = 1 + 0.115 + j0.1992 = 1.115 + j0.1992 \text{ pu} \\ &= \boxed{1.133 \text{ pu} \angle 10.13^\circ \quad (0.18 \text{ rad})}\end{aligned}\tag{128}$$

2. Comparison and Physical Interpretation:

$$X''_s < X'_s < X_s \iff 0.23 < 0.33 < 1.20 \text{ pu}\tag{129}$$

$$|E''_{af}| < |E'_{af}| < |E_{af}| \iff 1.13 < 1.20 < 1.91 \text{ pu}\tag{130}$$

$$\delta'' < \delta' < \delta \iff 10.1^\circ < 13.8^\circ < 33.0^\circ\tag{131}$$

Physical Mechanism of the Three Time Frames:

- **Subtransient Period ($t \approx 10 - 20 \text{ ms}$, first few cycles):** When a sudden electrical fault or disturbance occurs, flux linkages in both the **dampers windings** (amortisseur) and the rotor field winding cannot change instantaneously (Lenz's law). Heavy eddy currents are induced in the damper bars, trapping the flux in high-reluctance leakage paths outside the rotor body. Consequently, the equivalent reactance X''_s is minimal, and the initial fault current is very large.
- **Transient Period ($t \approx 0.1 - 1.0 \text{ s}$):** Due to the high resistance of the damper bars, the subtransient damper currents rapidly decay with time constant T''_d . Flux is now prevented from changing only by the main field winding, whose resistance is much smaller. The corresponding effective reactance is X'_s , and the internal voltage is $\bar{E}'_{af}$.
- **Steady-State Period ($t > 1 - 2 \text{ s}$):** After several seconds, all induced transient rotor currents have fully dissipated. The field winding carries only DC excitation current supplied by the exciter, and the effective reactance increases to its full steady-state value X_s (governed by stator leakage plus the full armature reaction mmf).

6 Practical Session 6: Voltage Instability in Power Systems

6.1 Exercises

1. Derivation of the P - V Characteristic of a Transmission Line⁶

Consider the transmission system represented in Figure 14. A load consuming an active power P and a reactive power Q is connected to bus 2. The voltage phasor at bus 2 is:

$$\bar{V} = V\angle\delta.$$

A synchronous generator is connected to bus 1 and regulates the voltage at that bus to:

$$\bar{E} = E\angle 0^\circ.$$

The transmission line between bus 1 and bus 2 is assumed to be lossless with series reactance X , and the reactive power limits of the generator are not considered.

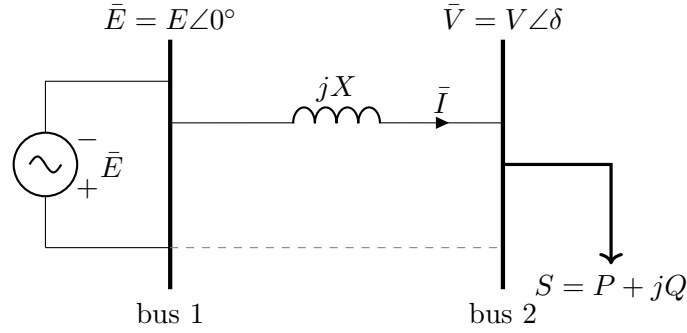


Figure 14: Single-line representation of a lossless transmission line feeding a load.

Based on these assumptions:

- Derive the mathematical relationship between the active power P consumed by the load and the receiving-end voltage magnitude V at bus 2, for a constant load power factor angle ϕ ($Q = P \tan \phi$).
- Derive the mathematical expression of the maximum active power $P_{\max}$ transmissible through the line.
- Identify and discuss the critical parameters to consider in order to transmit more power through the line.

⁶Exercise from Practical Session 6: Voltage instability in power systems.

6.2 Solutions

1. Derivation of the Relationship Between Load Active Power P and Voltage Magnitude V

Complex Power Flow Equations: The complex power S consumed by the load at bus 2 is:

$$S = P + jQ = \bar{V}\bar{I}^* \quad (132)$$

The line current flowing from bus 1 to bus 2 through the reactance jX is:

$$\bar{I} = \frac{\bar{E} - \bar{V}}{jX} \quad (133)$$

Substituting this current into the complex power expression:

$$S = \bar{V} \left(\frac{\bar{E} - \bar{V}}{jX} \right)^* = (V\angle\delta) \left(\frac{E - V\angle\delta}{jX} \right)^* \quad (134)$$

Using the complex conjugate identity $\left(\frac{1}{j}\right)^* = (-j)^* = j$:

$$\begin{aligned} S &= V\angle\delta \left(-j \frac{E - V\angle\delta}{X} \right)^* = V\angle\delta \left(j \frac{E - V\angle(-\delta)}{X} \right) \\ &= \frac{V}{X} \angle\delta [jE - jV(\cos\delta - j\sin\delta)] \\ &= \frac{V}{X} \angle\delta [-V\sin\delta + j(E - V\cos\delta)] \\ &= \frac{V}{X} (\cos\delta + j\sin\delta) [-V\sin\delta + j(E - V\cos\delta)] \end{aligned} \quad (135)$$

Expanding the product into real and imaginary parts:

Active Power P :

$$\begin{aligned} P &= \text{Re}\{S\} = \frac{V}{X} [\cos\delta(-V\sin\delta) + \sin\delta(E - V\cos\delta)] \\ &= \frac{V}{X} [-V\cos\delta\sin\delta + E\sin\delta - V\sin\delta\cos\delta] \\ &= \boxed{\frac{EV}{X} \sin(-\delta) = -\frac{EV}{X} \sin\delta} \end{aligned} \quad (136)$$

Taking the convention where $\delta < 0$ (load voltage lags the source voltage), let $\theta = -\delta > 0$:

$$P = \frac{EV}{X} \sin\theta \quad (137)$$

Reactive Power Q :

$$\begin{aligned} Q &= \text{Im}\{S\} = \frac{V}{X} [\cos\delta(E - V\cos\delta) - \sin\delta(-V\sin\delta)] \\ &= \frac{V}{X} [E\cos\delta - V\cos^2\delta + V\sin^2\delta] \\ &= \boxed{\frac{V}{X} (E\cos\delta - V)} \end{aligned} \quad (138)$$

Eliminating the Phase Angle δ : From equations (136) and (138):

$$\sin \delta = -\frac{PX}{EV} \quad (139)$$

$$\cos \delta = \frac{QX + V^2}{EV} \quad (140)$$

Applying the trigonometric identity $\sin^2 \delta + \cos^2 \delta = 1$:

$$\left(\frac{PX}{EV}\right)^2 + \left(\frac{QX + V^2}{EV}\right)^2 = 1 \quad (141)$$

Multiplying through by $(EV)^2$:

$$P^2X^2 + (QX + V^2)^2 = E^2V^2 \quad (142)$$

Expanding the second term:

$$P^2X^2 + Q^2X^2 + 2QXV^2 + V^4 = E^2V^2 \quad (143)$$

Grouping terms into a quadratic polynomial in V^2 :

$$\boxed{V^4 + (2QX - E^2)V^2 + X^2(P^2 + Q^2) = 0} \quad (144)$$

Solving for the Voltage Magnitude V : Let $y = V^2$. The quadratic equation is $y^2 + (2QX - E^2)y + X^2(P^2 + Q^2) = 0$. Applying the quadratic formula:

$$\begin{aligned} y &= \frac{-(2QX - E^2) \pm \sqrt{(2QX - E^2)^2 - 4X^2(P^2 + Q^2)}}{2} \\ &= \frac{E^2 - 2QX \pm \sqrt{4Q^2X^2 - 4E^2QX + E^4 - 4P^2X^2 - 4Q^2X^2}}{2} \\ &= \frac{E^2}{2} - QX \pm \sqrt{\frac{E^4}{4} - E^2QX - P^2X^2} \end{aligned} \quad (145)$$

Taking the square root to obtain the voltage magnitude $V = \sqrt{y}$:

$$\boxed{V = \sqrt{\frac{E^2}{2} - QX \pm \sqrt{\frac{E^4}{4} - E^2QX - P^2X^2}}} \quad (146)$$

For a load operating at a constant power factor $\cos \phi$, the reactive power is proportional to active power:

$$Q = P \tan \phi \quad (147)$$

where $\phi > 0$ for inductive (lagging) loads and $\phi < 0$ for capacitive (leading) loads. Substituting $Q = P \tan \phi$ into (146):

$$\boxed{V = \sqrt{\frac{E^2}{2} - X(P \tan \phi) \pm \sqrt{\frac{E^4}{4} - E^2X(P \tan \phi) - X^2P^2}}} \quad (148)$$

This equation describes the classical **P - V nose curve** of the transmission line. For each valid power $P < P_{\max}$, there are two mathematical voltage solutions:

- **Upper branch (+ sign):** High voltage, low current, physically stable operating point.
- **Lower branch (− sign):** Low voltage, high current, unstable operating point.

2. Maximum Transmissible Active Power $P_{\max}$

The maximum transmissible power $P_{\max}$ occurs at the **nose** (turning point / saddle-node bifurcation) of the P - V curve, where the two voltage solutions merge into a single unique point. This condition is reached when the discriminant Δ under the inner square root vanishes:

$$\Delta = \frac{E^4}{4} - E^2 X (P_{\max} \tan \phi) - X^2 P_{\max}^2 = 0 \quad (149)$$

Rearranging into standard quadratic form for $P_{\max}$:

$$X^2 P_{\max}^2 + (E^2 X \tan \phi) P_{\max} - \frac{E^4}{4} = 0 \quad (150)$$

Dividing by X^2 :

$$P_{\max}^2 + \left(\frac{E^2 \tan \phi}{X} \right) P_{\max} - \frac{E^4}{4X^2} = 0 \quad (151)$$

Using the quadratic formula and selecting the positive physical root:

$$\begin{aligned} P_{\max} &= \frac{-\left(\frac{E^2 \tan \phi}{X}\right) + \sqrt{\left(\frac{E^2 \tan \phi}{X}\right)^2 - 4(1)\left(-\frac{E^4}{4X^2}\right)}}{2} \\ &= \frac{1}{2} \left[-\frac{E^2 \tan \phi}{X} + \sqrt{\frac{E^4 \tan^2 \phi}{X^2} + \frac{E^4}{X^2}} \right] \\ &= \frac{E^2}{2X} \left[-\tan \phi + \sqrt{\tan^2 \phi + 1} \right] \end{aligned} \quad (152)$$

Using the trigonometric identity $1 + \tan^2 \phi = \sec^2 \phi = \frac{1}{\cos^2 \phi}$:

$$\sqrt{1 + \tan^2 \phi} = \frac{1}{\cos \phi} \quad (153)$$

Substituting $\tan \phi = \frac{\sin \phi}{\cos \phi}$:

$$\begin{aligned} P_{\max} &= \frac{E^2}{2X} \left[-\frac{\sin \phi}{\cos \phi} + \frac{1}{\cos \phi} \right] \\ &= \boxed{\frac{E^2}{2X} \left(\frac{1 - \sin \phi}{\cos \phi} \right)} \end{aligned} \quad (154)$$

Critical Voltage at the Limit: At $P = P_{\max}$, the inner discriminant is zero, so:

$$V_{\text{crit}} = \sqrt{\frac{E^2}{2} - X Q_{\text{crit}}} = \sqrt{\frac{E^2}{2} - X P_{\max} \tan \phi} \quad (155)$$

Substituting $P_{\max}$ from (154):

$$\begin{aligned} V_{\text{crit}}^2 &= \frac{E^2}{2} - X \left[\frac{E^2}{2X} \left(\frac{1 - \sin \phi}{\cos \phi} \right) \right] \frac{\sin \phi}{\cos \phi} \\ &= \frac{E^2}{2} \left[1 - \frac{\sin \phi - \sin^2 \phi}{\cos^2 \phi} \right] = \frac{E^2}{2} \left[\frac{\cos^2 \phi - \sin \phi + \sin^2 \phi}{\cos^2 \phi} \right] \\ &= \frac{E^2}{2} \left[\frac{1 - \sin \phi}{1 - \sin^2 \phi} \right] = \frac{E^2}{2(1 + \sin \phi)} \end{aligned} \quad (156)$$

$$\boxed{V_{\text{crit}} = \frac{E}{\sqrt{2(1 + \sin \phi)}}} \quad (157)$$

For a purely resistive load ($\phi = 0$):

$$P_{\max} = \frac{E^2}{2X}, \quad V_{\text{crit}} = \frac{E}{\sqrt{2}} \approx 0.707E$$

3. Key Parameters Influencing the Maximum Transmissible Power

From equation (154), three primary parameters determine the transmission limit:

1. **Sending-End Voltage Magnitude (E):** The maximum power increases quadratically with the sending-end voltage:

$$P_{\max} \propto E^2$$

Operating transmission grids at higher voltage levels (e.g., 400 kV or 765 kV) drastically increases the capacity of corridors.

2. **Line Series Reactance (X):** The maximum transmissible power is inversely proportional to the line reactance:

$$P_{\max} \propto \frac{1}{X}$$

To increase capacity on an existing corridor:

- Install **series capacitor banks** (series compensation) to reduce net line reactance: $X_{\text{eff}} = X_{\text{line}} - X_C$.
 - Construct parallel circuits or bundle multiple sub-conductors per phase to decrease line inductance.
3. **Load Power Factor ($\cos \phi$ and Angle ϕ):** The factor $\frac{1-\sin \phi}{\cos \phi}$ depends strongly on the power factor:

- **Inductive loads ($\phi > 0$, lagging pf):** $\sin \phi > 0 \implies \frac{1-\sin \phi}{\cos \phi} < 1$, reducing $P_{\max}$ significantly and worsening voltage drop.
- **Capacitive loads ($\phi < 0$, leading pf):** $\sin \phi < 0 \implies \frac{1-\sin \phi}{\cos \phi} > 1$, raising the maximum power limit and supporting voltage.

This underscores why shunt capacitors, STATCOMs, and SVCs are installed near load centers: by providing local reactive power support, they unload the line from transmitting reactive power, effectively increasing the active power transfer capability and preventing voltage collapse.